

P S Mithul Sourav

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EDUCATION

National Institute of Technology Karnataka (NITK), Surathkal
B.Tech. in Mechanical Engineering; CGPA: 7.37/10

Aug 2023 – Jun 2027
Surathkal, India

RESEARCH INTERESTS

I am broadly interested in experimental and field robotics, particularly rough-terrain mobile robots, and in the dynamics, vibration, sensing, control, and autonomy of mechatronic systems.

RELEVANT EXPERIENCE

Vayu Aerospace Pvt. Ltd.

Jun 2026 – Jul 2026
Bengaluru, India

Product Intern | UAV Vibration Analysis & Isolation

- Mapped how motor and airframe vibration reached the flight-controller IMU, which helped frame the mount as a vibration-path problem rather than just a bracket problem.
- Modeled the rigid, elastomer-isolated, and suspended mounts in ANSYS and compared natural frequencies, mode shapes, and response to understand how each support changed the vibration path.
- Integrated the shortlisted mounts into hardware and checked CAD fit, clearances, cable slack, fasteners, and controller orientation so the real assembly preserved the intended isolation path.
- Tested the assembled mounts through ground motor runs and analyzed IMU time histories, RMS levels, and FFT spectra in MATLAB, then combined those results with the simulation and integration issues to support the elastomer-isolated tray.

ONGOING PROJECTS

Design and Development of a Tensegrity Based Variable Stiffness Joint

May 2026 – Present

- Modeled a tensegrity joint to study how geometry and internal loading can change the stiffness available to a wearable or exoskeleton joint.
- Analyzed force output and joint stiffness in MATLAB while varying geometry, external loading, and internal force.
- Isolated the design changes that have the strongest effect on force and stiffness instead of varying many dimensions at once.
- Narrowed the study toward a small set of configurations that are worth carrying into a physical exoskeleton-oriented prototype.

Adaptive Suspension and Vibration-Aware Control for a Rough-Terrain Rover

Sep 2026 – Present

- Designed a six-wheel rough-terrain rover around one question: can the suspension change stiffness before vibration forces the rover to slow down?
- Designed soft, medium, and stiff suspension settings with mechanical locking and a separate load path so the rover remains supported if the stiffness-changing mechanism fails.
- Defined a vibration-aware control loop using IMU and wheel-encoder data to detect sustained vibration, change stiffness, verify the new state, and reduce speed only when reconfiguration is not enough.
- Structured the validation from a single suspension unit to full rover trials, comparing fixed-stiffness, speed-only, stiffness-only, and combined control.

Off-Road Leaf-Collection Robot

Jul 2026 – Present

- Defined the core pickup problem as collecting dry and wet leaves on uneven ground without pulling in too much soil.
- Decomposed the robot into locomotion, pickup, transfer, storage, and terrain-following systems so each part can be tested before committing to the full design.
- Screened pickup concepts against space, manufacturability, and integration limits, then built the BOM around the most practical options.
- Defined tests for traction, pickup efficiency, soil rejection, and endurance so the next design change comes from data rather than preference.

SELECTED COMPLETED PROJECTS

Development of Actuated Knee Assistance System

Oct 2025 – Sep 2026

- Developed a passive spring-assisted knee concept to study how spring placement and joint geometry change the assistance available through the motion.
- Modeled and refined the mechanism in SolidWorks, then used ANSYS and interference checks to identify layouts that could move through the intended flexion range.
- Extended the passive concept into an actuated brace that tracks knee angle and adds controlled cable-and-spring assistance while the user extends actively.
- Converted the final design into a procurement-ready system with defined drive, sensing, power, manual release, and bench tests; our team placed in the top 5 of 70 teams nationwide in the Incubate X Prosthetic Challenge.

Reaction-Wheel Microvibration Prediction with FEM and Machine Learning

Aug 2026 – Oct 2026

- Modeled reaction-wheel microvibration with modal and harmonic FEM to trace the disturbance from the wheel mount through the satellite panel to the camera interface.
- Identified a strong camera response near 5,800 rpm and refined the mesh to test whether the peak was numerical; the amplitude changed by only 0.33% between the compared meshes.
- Focused the analysis on the camera interface so the study tracked what the payload location experiences, not only where the panel resonates.
- Structured a 24-design parameter sweep and held-out regression check to test when a faster surrogate model can follow the FEM response and where it starts to fail near resonance.

Uncertainty-Aware Navigation for an Indoor Mobile Robot

Sep 2025 – Dec 2025

- Implemented a differential-drive simulator with A* path planning, heading control, wheel encoders, an IMU, and extended Kalman filter localization.
- Injected encoder noise, IMU drift, yaw bias, and wheel slip so the robot had to navigate with imperfect state estimates.
- Implemented confidence-aware speed and recovery logic so the robot could respond when the state estimate became unreliable.
- Evaluated the navigation stack through Monte Carlo trials, tracking collisions, localization error, recovery events, success, and time to goal to understand failure modes, not just final success.

LEADERSHIP AND RECOGNITION

Indian Society for Technical Education (ISTE), NITK

Sep 2024 – Present

Secretary

NITK, Surathkal

- Led sponsor outreach, budgeting, vendor coordination, and logistics for a technical event with 500+ participants, taking responsibility for both planning and execution.
- Secured three sponsors and coordinated across teams as budgets, vendors, and timelines changed.

NH66 Fund, P&L Club

Jan 2025 – Apr 2026

Fund Manager

NITK, Surathkal

- Managed strategy for an INR 150K student fund and mentored three junior analysts on how to structure and defend an investment view.
- Organized two recruitment drives and a training workshop so new analysts could learn the process and contribute faster.

Global Case Competition at Harvard

Feb 2026 – Mar 2026

Top 50 globally

Team of four

- Analyzed the European defence landscape in a four-member team and turned a broad case into a focused set of recommendations.
- Reached the global Top 50 in the 2026 competition by working through evidence, trade-offs, and the reasoning behind each recommendation.

OTHER SELECTED EXPERIENCE

Thinkify Labs

May 2025 – Aug 2025

Product and Strategy Intern

Bengaluru, India

- Developed an AI-based lead-qualification workflow that turned profile and email information into a repeatable screening process.
- Defined routing logic so useful prospects reached the right sales representatives instead of every lead being handled the same way.
- Improved the workflow into a structured, testable process that contributed to a reported 25% improvement in operational efficiency.

TECHNICAL SKILLS

Software: MATLAB; ANSYS Mechanical; SolidWorks.

Programming: Python; MATLAB; C; C++.

Languages: English; Hindi; Tamil; French (basic).